

1. DIET RECOMMENDATION-DISEASE

AIM

To write a Prolog program to suggest a suitable diet based on a given disease using facts and rules.

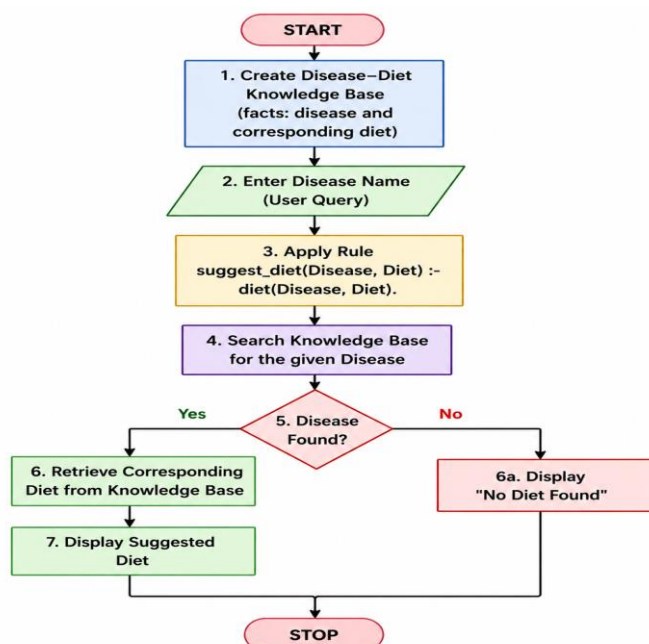
OBJECTIVE

- To create a disease–diet knowledge base in Prolog.
- To associate diseases with appropriate diet types.
- To use Prolog rules to retrieve diet suggestions.
- To understand knowledge representation and logical inference.

ALGORITHM

1. Start.
2. Define diseases and their corresponding diet types as facts.
3. Define the suggest_diet rule.
4. Enter the disease as a query.
5. Search the database for the matching disease.
6. If a match is found, retrieve the corresponding diet.
7. Display the suggested diet.
8. Stop.

FLOWCHART



OUTPUT

diet(diabetes, low_sugar).

diet(hypertension, low_salt).

diet(obesity, low_calorie).

diet(anemia, iron_rich).

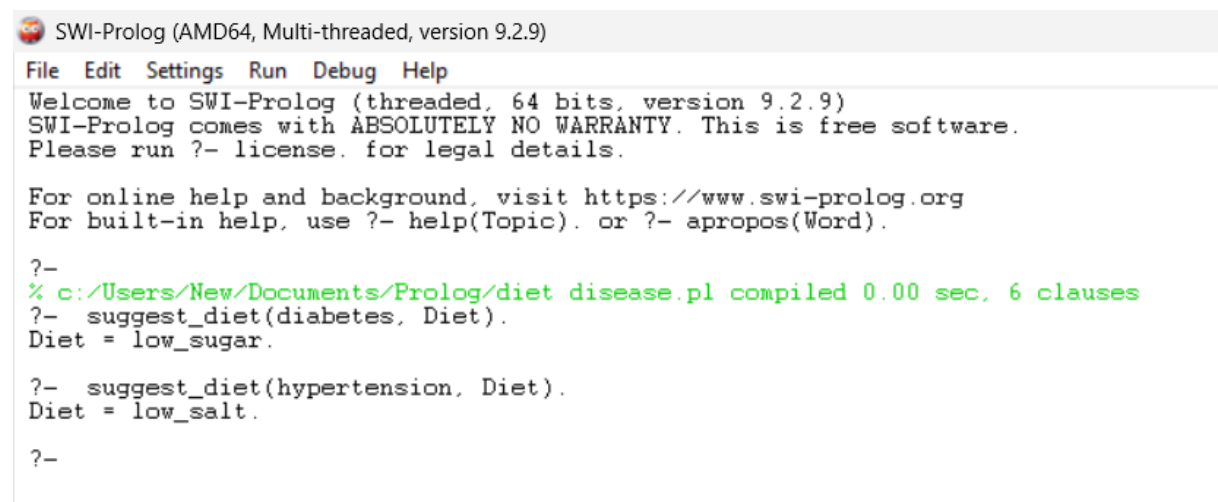
suggest_diet(Disease, Diet) :-

diet(Disease, Diet).

QUERY

?- suggest_diet(diabetes, Diet).

OUTPUT



```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/New/Documents/Prolog/diet disease.pl compiled 0.00 sec, 6 clauses
?- suggest_diet(diabetes, Diet).
Diet = low_sugar.

?- suggest_diet(hypertension, Diet).
Diet = low_salt.

?-
```

RESULT

Thus, the Prolog program was successfully implemented to suggest a diet based on the given disease, and the appropriate diet was retrieved using Prolog queries.